Chapter test

3 Cellular organization

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Robert Hooke was the first scientist to observe cells using a light microscope. His study led to

   A. the discovery of bacteria as pathogens.
   B. the discovery of sub-cellular structures.
   C. the development of the cell theory.
   D. the invention of electron microscopes. ☐

2. A student uses a light microscope to examine a specimen of cells. If he wants to observe more cells within the field of view, he should

   A. stain the cells with a suitable dye.
   B. change to an objective with a lower magnification.
   C. turn the coarse knob to reduce the distance between the specimen and the objective.
   D. adjust the iris diaphragm. ☐

3. Which cell structure(s) can be seen with a light microscope at 400x magnification?

   (1) cell wall
   (2) mitochondrion
   (3) chloroplast

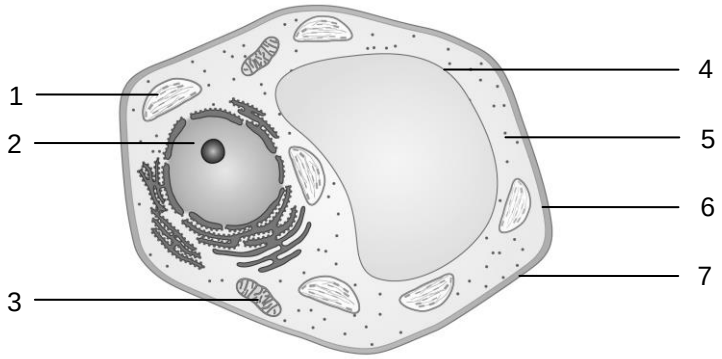
   A. (1) only
   B. (1) and (2) only
   C. (1) and (3) only
   D. (1), (2) and (3) ☐

4. Which of the following organelles is ***not*** bound by a double membrane?

   A. mitochondrion
   B. chloroplast
   C. nucleus
   D. vacuole ☐

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Directions: Questions 5 to 7 refer to the diagram below, which shows the structure of a plant cell.



5. Which structure(s) can be used to distinguish the plant cell from an animal cell?

   A. 1 and 7 only
   B. 3 and 7 only
   C. 1, 4 and 7 only
   D. 3, 4 and 7 only ☐

6. Which structure(s) is / are differentially permeable?

   A. 6 only
   B. 4 and 6 only
   C. 6 and 7 only
   D. 4, 6 and 7 only ☐

7. Which structures of the cell carry out the functions below?

	Synthesizes carbohydrates	***Prevents the entry of excess water***
A.	1	6
B.	1	7
C.	2	6
D.	2	7 ☐

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Directions: Questions 8 and 9 refer to the table below, which shows some characteristics of four cells.

Cell	Endoplasmic reticulum	Ribosome	Cell wall	Flagellum
P	✓	✓	✓	×
Q	✓	✓	×	×
R	✓	✓	×	✓
S	×	✓	✓	✓

★ 8. Which cell is most likely a bacterial cell?

A. P
B. Q
C. R
D. S ☐

9. Cell P is a eukaryotic cell because

(1) it has endoplasmic reticulum.
(2) it has ribosomes.
(3) it has a cell wall.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

10. What are the levels of organization in an organism?

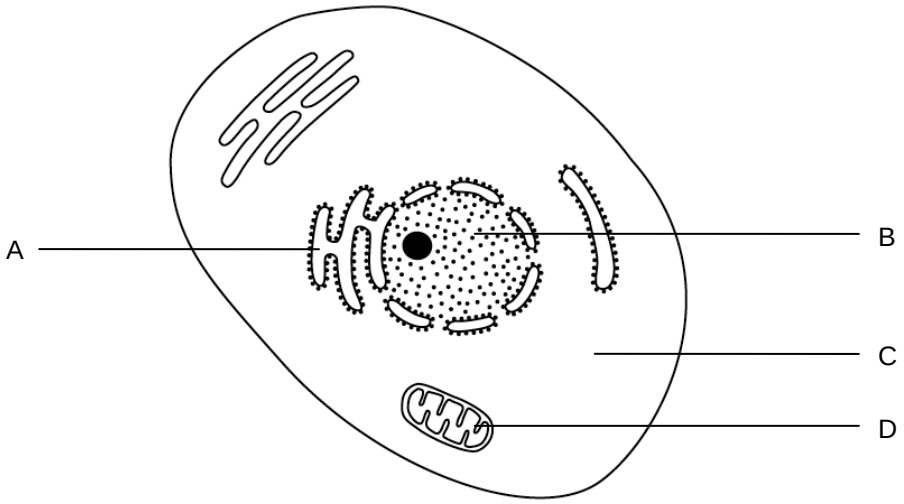
A. organism → cells → organs → tissues → systems
B. tissues → cells → organs → systems → organism
C. cells → organs → systems → tissues → organism
D. cells → tissues → organs → systems → organism ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	3.1	3.1	3.3	3.3	3.3	3.3	3.3	3.4	3.4	3.5

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B. Problem solving (20 marks)

★ 11. The diagram below shows an animal cell.



(a) Complete the table below to give names and functions of structures A, B and C. (6 marks)

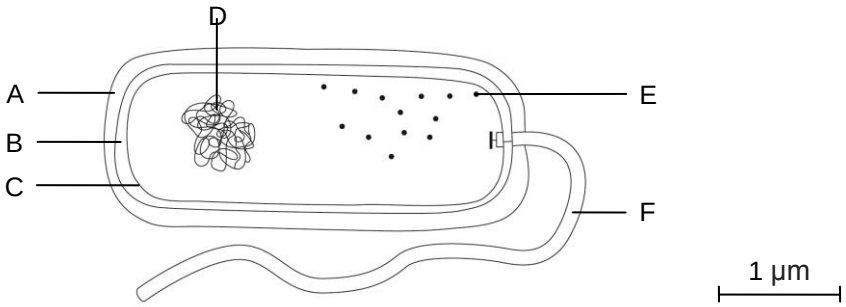
Structure	Name	Function
A		
B		
C		

(b) Suggest ***one*** type of human cell which contains a large number of structure D. Explain the significance of this type of cell having such a large number of structure D. (2 marks)

(Total: 8 marks)

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12. The diagram below shows a prokaryotic cell.



(a) Using the scale bar, determine the actual length of this prokaryotic cell. (1 mark)

(b) State ***two*** differences between a prokaryotic cell and a eukaryotic cell. (2 marks)

(c) Using the letters provided, state ***two*** structures visible in the diagram that are ***not*** present in all prokaryotic cells. (2 marks)

(d) What is the difference between the compositions of structure B of the prokaryotic cell and that of a plant cell? (1 mark)

(Total: 6 marks)

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13. (a) Discuss the cell theory in relation to the characteristics of unicellular organisms. (3 marks)

(b) The human body is made up of many cells working together at different levels to perform different functions.

(i) Why can blood be classified as a tissue? (1 mark)

(ii) What is an organ? (1 mark)

(iii) What is a system? (1 mark)

(Total: 6 marks)

C. Exam challenge (10 marks)

★ 14. (a) A student used a light microscope to examine a specimen of onion epidermal cells. The eyepiece used has a magnification of 15x and the objective has a magnification of 10x. Calculate the total magnification of the microscope. (2 marks)

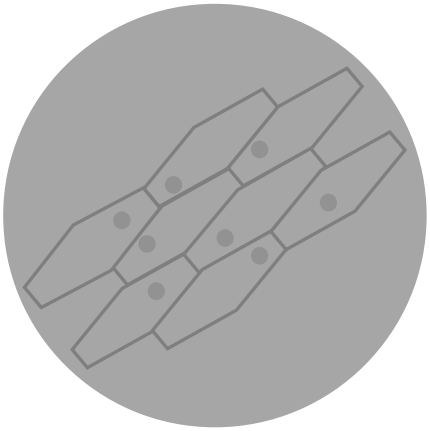
(b) The cells were stained before being examined.

(i) Explain why this is necessary. (1 mark)

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(ii) Suggest a suitable dye for staining plant cells. (1 mark)

(c) The diagram below shows the field observed by the student.



Describe how the student should adjust the microscope in order to get a brighter view. (2 marks)

(d) For each case below, state whether a light microscope or an electron microscope should be used for studying the specimen and give a reason for your answer.

(i) Observing the detailed structure of a mitochondrion (2 marks)

(ii) Observing how an *Amoeba* feed (2 marks)

(Total: 10 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	B	3.	C	4.	D	5.	C
6.	B	7.	B	8.	D	9.	A	10.	D

(1 mark each)

B. Problem solving (20 marks)

11. (a)

Structure	Name	Function
A	rough endoplasmic reticulum (1)	transports proteins (1)
B	nucleus (1)	controls cell activities (1)
C	cytoplasm (1)	provides a medium for many chemical reactions to take place (1)

(b) Any ***one set*** of the following: (2)

- sperm (1)
  to provide energy for the sperms to swim towards the egg (1)
- muscle fibre (1)
  to provide energy for muscle contraction (1)

(Accept other reasonable answers.)

(Total: 8 marks)

12. (a) 3.5 μm (1)

(b) The DNA of a prokaryotic cell lies free in the cytoplasm, while the DNA of a eukaryotic cell is enclosed in the nuclear membrane. (1)
A eukaryotic cell has membrane-bounded organelles, while a prokaryotic cell does not. (1)

(c) A, F (1, 1)

(d) Structure B (the cell wall) of the prokaryotic cell does not contain cellulose as a plant cell does. (1)

(Total: 6 marks)

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13. (a) All living organisms are made up of one or more cells. A unicellular organism is made up of a cell. (1)
The cell is the basic unit of life. The cell of a unicellular organism can carry out all the functions of life. (1)
Cells arise from pre-existing cells. Unicellular organisms reproduce by cell division. (1)

(b) (i) Blood consists of different types of blood cells working together for the transport of substances around the body. (1)

(ii) An organ consists of different tissues grouped together to perform particular functions. (1)

(iii) A system consists of different organs and tissues working together to perform particular functions. (1)

(Total: 6 marks)

C. Exam challenge (10 marks)

14. (a) 15×10 (1)
$= 150x$ (1)

(b) (i) To add contrast to the specimen for easier observation (1)

(ii) iodine solution (1)

(c) Adjust the mirror to reflect more light to the specimen. (1)
Adjust the iris diaphragm to allow more light to pass through the condenser. (1)

(d) (i) Electron microscope (1)
because it has high magnifying power for viewing sub-cellular structures / the magnification of a light microscope is not high enough. (1)

(ii) Light microscope (1)
because it can be used for viewing live specimen / specimens for examination by electron microscopes must be dead. (1)

(Total: 10 marks)

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